

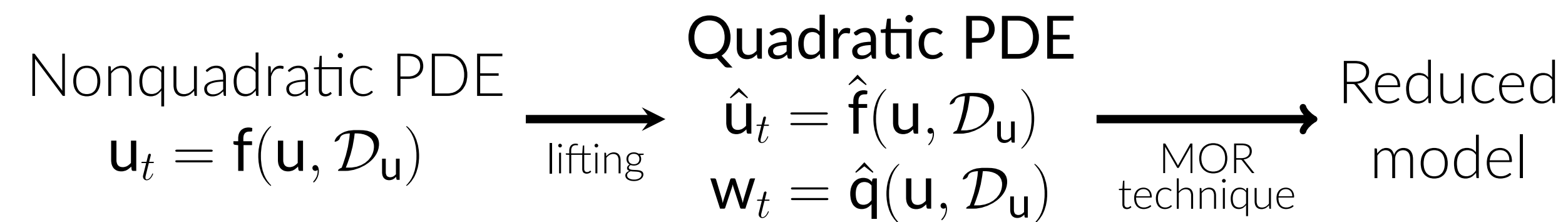
Discovering Quadratic Representations of PDEs: Algorithms and Software

Albani Olivieri¹ Gleb Pogudin² Boris Kramer¹

¹Department of Mechanical and Aerospace Engineering, University of California San Diego ²LIX, CNRS, École Polytechnique, Institute Polytechnique de Paris

Background and Motivation

- Nonpolynomial and nonquadratic PDEs describe complex dynamical processes in science and engineering.
- Lifting PDE systems to quadratic form before applying MOR techniques can yield better results, see [GU2011, LIFTING2019, LIFT&LEARN].



- No algorithmic tool exists to do this lifting directly for PDEs.

Research Problem

Design an algorithm that finds quadratic liftings/transformations for PDEs.

Example: Quadratization for a PDE

- Consider a nonquadratic PDE for $u(t, x)$:

$$u_t = u_x u^2. \quad (1)$$

- We introduce a new variable with its partial derivative in x :

$$y = u^2 \Rightarrow y_x = 2u_x u.$$

- We re-write the original PDE and add a differential equation corresponding to the new variable:

$$u_t = u_x y$$

$$y_t = y_x y.$$

The set $\{u^2\}$ is called a monomial quadratization for Eq. (??).

Research Problem

Can we find monomial quadratizations quickly and ensure optimality, i.e., that we add the least number of variables?

Algorithm QuPDE (Quadratization for Partial Differential Equations)

A) Which lifting variables should the algorithm suggest?

Monomial decomposition: How many ways can we divide a set of elements of size n into two subsets?

$$u_t = u_x^2 u$$

$$\begin{array}{c} | \\ \hline u, u_x^2 \quad u_x u, u_x \quad u_x^2 u, 1 \end{array}$$

The new variable candidates would be $\{u_x^2, u_x u, u_x^2 u\}$.

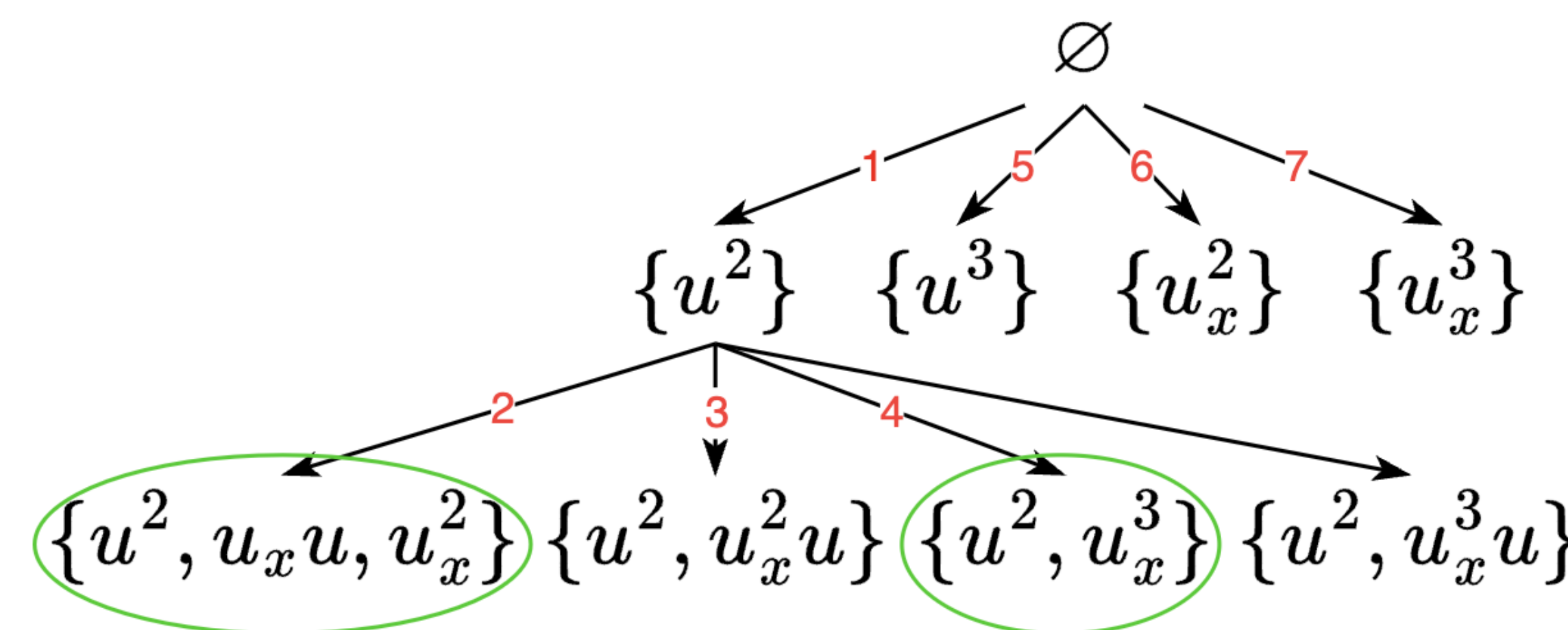
B) How can we automate the search for the best quadratization?

1. Branch-and-Bound (B&B)

For each node (set) that is not a quadratization, we add as children all the subproblems that come from the transformed PDE of the parent node. If we consider the example PDE:

$$u_t = u_x^3 + u^3, \quad (2)$$

its search tree for finding a quadratization is:

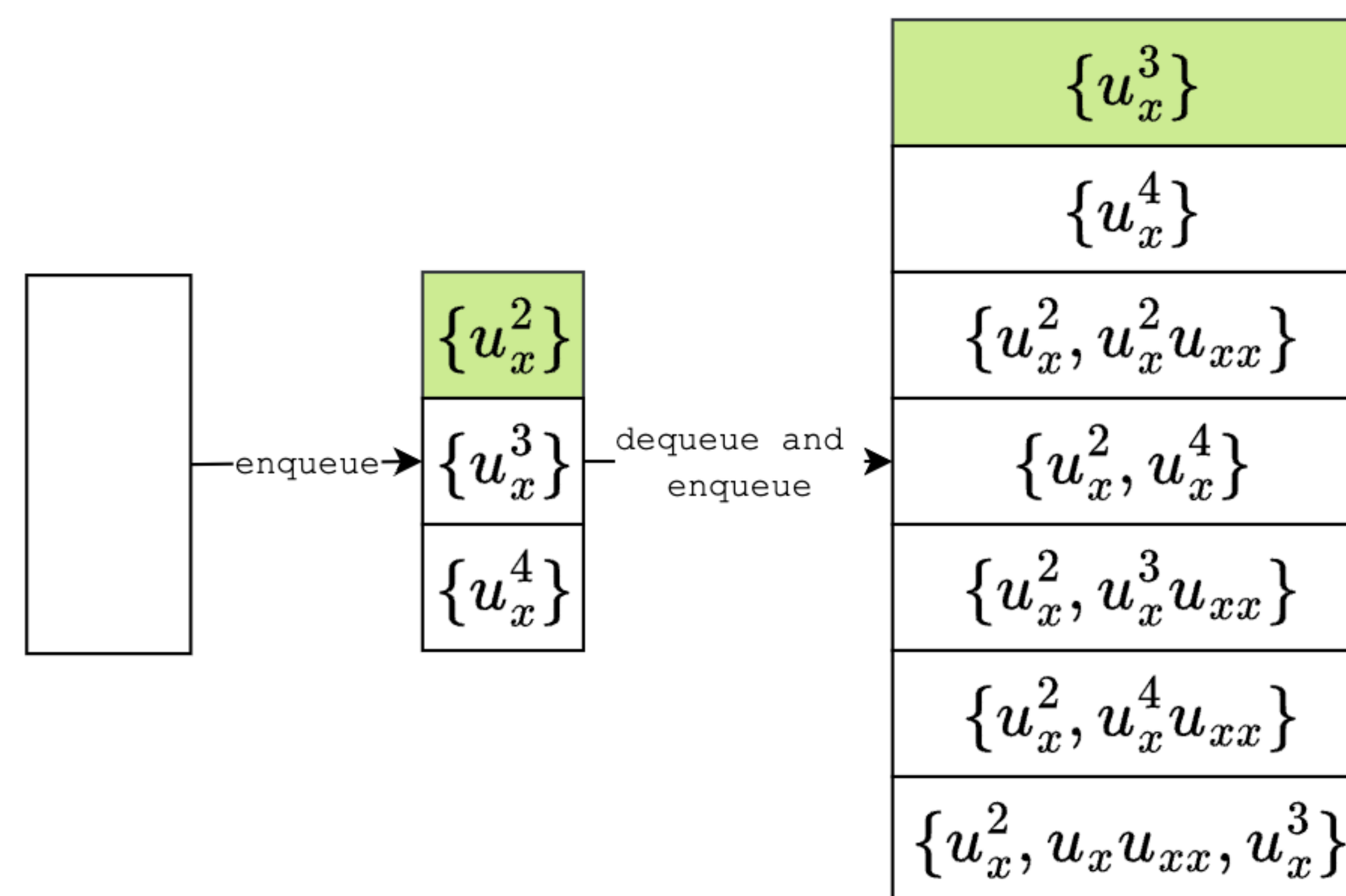


2. Incremental Nearest Neighbor (INN)

In each iteration, we check whether the best set stored so far is a quadratization. If not, we store all its subproblems. For the PDE:

$$u_t = u_x^4 \quad (3)$$

this algorithm would do the following:



C) How can we check if a set is a quadratization?

Consider the sets:

- $\mathcal{V}$: The set of variables that are a potential quadratization plus the PDE's original variables.

- $\mathcal{V}^2$: $\{v_1 v_2 | v_1, v_2 \in \mathcal{V}\}$

Goal: If a set is a quadratization for a PDE, then each equation can be written as a linear combination of elements in $\mathcal{V}^2$.

How?: With $\mathcal{V}^2$ expressed as a reduced row-echelon form (RREF) matrix, we try to reduce each nonquadratic polynomial right-hand side with respect to $\mathcal{V}^2$ (Gaussian elimination).

Results for both search algorithms: B&B and INN

PDE system	New variables	Time B&B (s)	Time INN (s)
Solar wind model	$1/u$	0.01 ± 0.0001	0.02 ± 0.0002
Allen-Cahn equation	u^2	0.04 ± 0.03	0.04 ± 0.0002
mKdV equation	u^2	0.04 ± 0.03	0.03 ± 0.004
Schlögl model	u^2	0.05 ± 0.03	0.04 ± 0.0003
Euler equations	$1/\rho$	0.11 ± 0.03	0.10 ± 0.001
Dym equation	$u^3, u_x^2 u$	0.64 ± 0.04	1.16 ± 0.01
Brusselator system	u^2, uv	0.82 ± 0.03	0.81 ± 0.005
Nonlinear heat equation	$u^3, u_x u, u^5$	3.02 ± 0.13	1.66 ± 0.004
Tubular reactor	$uv, v^2, v^2 u, v^3$	1874.77	2284.24

Comparing QuPDE (for PDEs) with QBee (for ODEs)

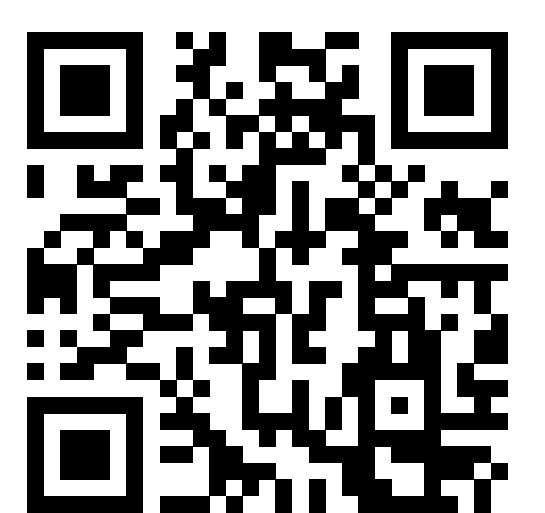
PDE system	QBee [bychkov2023exact]		QuPDE	
	Time (s)	Quad order	Time (s)	Quad order
Solar wind model	0.34	4	0.01	1
Allen-Cahn equation	0.07	1	0.04	1
Tubular reactor model	0.68	4	1874.77	4

Summary and Future Work

We developed an algorithm that explores a combinatorial tree of possible transformations and outputs a minimal set of variables that effectively quadratize a PDE system. Ideas for future work include:

- Develop a polynomialization algorithm.
- Improve the efficiency of the algorithm in terms of time.
- Reduce the number of equations of the resulting quadratic PDEs.

Link to software repository →



Acknowledgements

This work is supported by NSF-CMMMI award 2144023.

References